

## PRESSURE MEASUREMENT DEVICES

The pressure of a fluid is measured by the following devices:

1. Manometers
2. Mechanical Gauges.

### Manometers

A manometer is an instrument that uses a column of liquid to measure pressure, although the term is currently often used to mean any pressure instrument. Manometers are defined as the devices used for measuring the pressure at a point in a fluid by balancing the column of fluid by the same or another column of the fluid. They are classified as:

- (a) Simple Manometers,
- (b) Differential Manometers.

### Simple Manometers

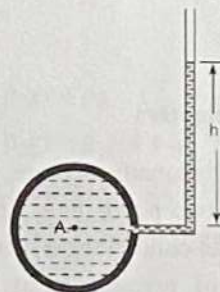
A simple manometer consists of a glass tube having one of its ends connected to a point where pressure is to be measured and other end remains open to atmosphere. Common types of simple manometers are:

1. Piezometer
2. U-tube Manometer
3. Single Column Manometer.

### Piezometer

It is the simplest form of manometer used for measuring gauge pressures. One end of this manometer is connected to the point where pressure is to be measured and other end is open to the atmosphere as shown in Figure. The rise of liquid gives the pressure head at that point. If at a point A, the height of liquid say water is  $h$  in Piezometer tube, then pressure at A

$$= \rho \times g \times h \frac{\text{N}}{\text{m}^2}.$$



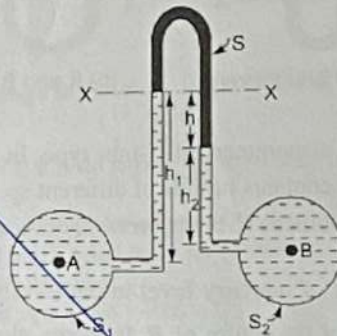
### U-tube Manometer

It consists of glass tube bent in U-shape, one end of which is connected to a point at which pressure is to be measured and other end remains open to the atmosphere as shown in Figure.. The tube generally contains mercury or any other liquid whose specific gravity is greater than the specific gravity of the liquid whose pressure is to be measured.



### Inverted U-tube differential manometer

It consists of an inverted U-tube, containing a light liquid. The two ends of the tube are connected to the points whose difference of pressure is to be measured. It is used for measuring difference of low pressures. Figure shows an inverted U-tube differential manometer connected to the two points A and B. Let the pressure at A is more than the pressure at B.



Let  
 $h_1$  = Height of liquid in left limb below the datum line X-X  
 $h_2$  = Height of liquid in right limb  
 $h$  = Difference of light liquid  
 $\rho_1$  = Density of liquid at A  
 $\rho_2$  = Density of liquid at B  
 $\rho_s$  = Density of light liquid  
 $p_A$  = Pressure at A  
 $p_B$  = Pressure at B.

Taking X-X as datum line. Then pressure in the left limb below X-X  
 $= p_A - \rho_1 \times g \times h_1$ .

Pressure in the right limb below X-X  
 $= p_B - \rho_2 \times g \times h_2 - \rho_s \times g \times h$

Equating the two pressure

$$p_A - \rho_1 \times g \times h_1 = p_B - \rho_2 \times g \times h_2 - \rho_s \times g \times h$$

or

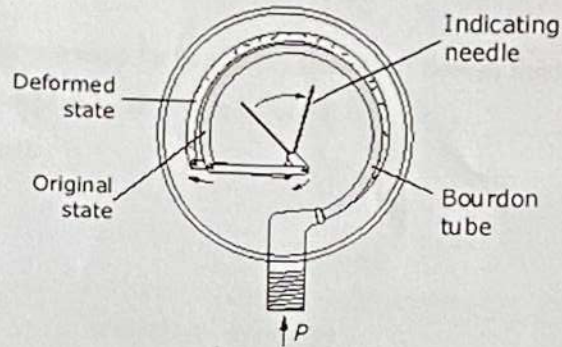
$$p_A - p_B = \rho_1 \times g \times h_1 - \rho_2 \times g \times h_2 - \rho_s \times g \times h.$$

### Bourdon Gage

The Bourdon Gauge is fitted with a transparent dial, which lets you see the internal workings of the gauge. The gauge consists of a thin walled closed ended tube which is oval in cross section. This tube is bent through an angle of about  $270^\circ$  along its long axis. The open end of the tube is welded to a hollow mounting block which allows the pressurized fluid to reach the tube. This causes the pressure from the source to be transmitted directly to the inside of the bourdon tube. The applied pressure causes the oval tube to become rounder (since a round cross section has the maximum area for a given circumference). As it becomes rounder, the bourdon tube tends to uncurl which causes its free end to move. A system of linkages and levers transmits this motion to the gauge needle which moves over the scale. The Bourdon tube is usually an elliptical cross-section tube having a 'C' shape configuration. When pressure is admitted to the inside of the tube, an elastic deformation results, which tries

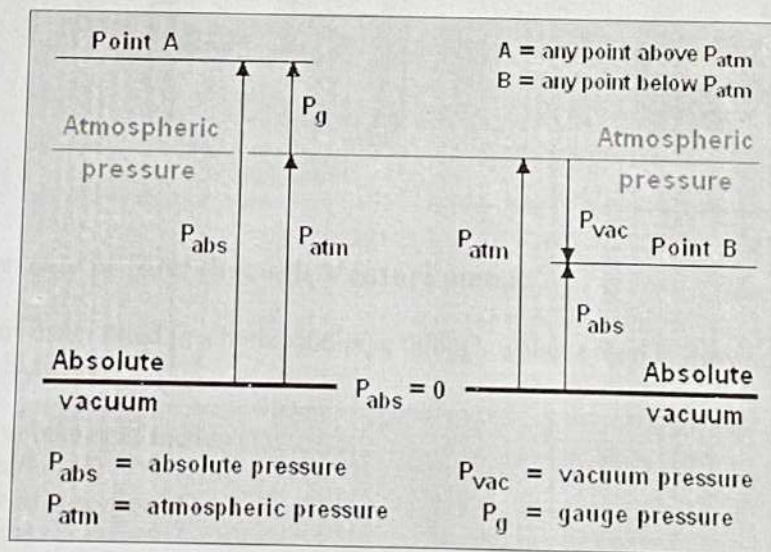


to straighten up the tube. One end of the tube is fixed and the other end is free to move and is connected to a spring-loaded linkage, which amplifies the displacement and transforms it into an angular rotation of the pointer.



Bourdon Gauge

Pressure of a fluid is the normal force exerted by a fluid on a unit area. The pressure designated will be either an absolute pressure or a gauge pressure. Absolute pressure is measured relative to a perfect vacuum (absolute zero pressure), whereas gauge pressure is measured relative to the local atmospheric pressure. Absolute pressures are positive, but gauge pressure can be either positive (above atmospheric pressure) or negative (below atmospheric pressure) as shown in figure



Graphical Representation of Gage & Absolute Pressure

|                       | kPa      | mm Hg    | millibar             | in H <sub>2</sub> O | PSI       |
|-----------------------|----------|----------|----------------------|---------------------|-----------|
| 1 atm                 | 101.325  | 760.000  | 1013.25              | 406.795             | 14.6960   |
| 1 kPa                 | 1.000    | 7.50062  | 10.000               | 4.01475             | 0.145038  |
| 1 mm Hg               | 0.133322 | 1.000    | 1.33322              | 0.535257            | 0.0193368 |
| 1 millibar            | 0.1000   | 0.750062 | 1.000                | 0.401475            | 0.0145038 |
| 1 in H <sub>2</sub> O | 0.249081 | 1.86826  | 2.49081              | 1.000               | 0.0361    |
| 1 PSI                 | 6.89473  | 51.7148  | 68.9473              | 27.6807             | 1.000     |
| 1 mm H <sub>2</sub> O | 0.009806 | 0.07355  | $9.8 \times 10^{-5}$ | 0.03937             | 0.0014223 |

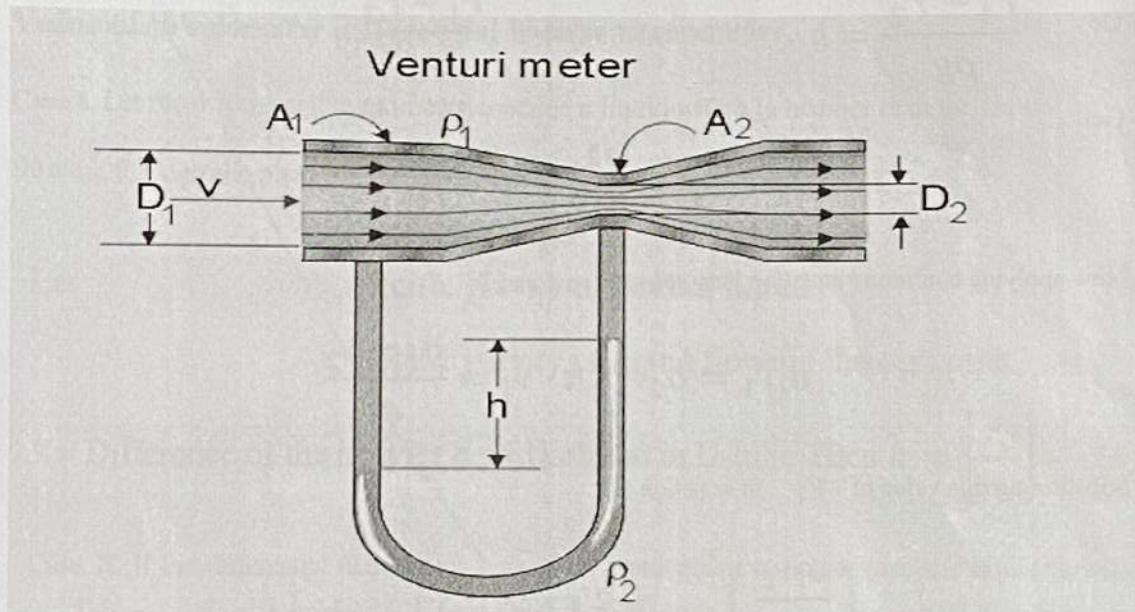


## FLOW MEASUREMENT DEVICES

### Venturi meter

A Venturi meter is a device used for measuring the rate of flow of fluid flowing through a pipe. It consists of three parts as shown in Figure 1:

- A short converging part
- Throat
- Diverging part



### Expression for rate of flow through Venturi meter:

Consider a Venturi meter fitted in a horizontal pipe through which a fluid is flowing (say water), as shown in figure

Let  $d_1$  = diameter at inlet or at section (1)

$P_1$  = pressure at section (1)

$v_1$  = velocity of fluid at section (1)

$a$  = area at section (1)

$d_2, p_2, v_2, a_2$ , are corresponding values at section (2).

Applying Bernoulli's equation at sections (1) and (2), we get

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

As pipe is horizontal, hence

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} \quad \text{Or} \quad \frac{P_1 - P_2}{\rho g} = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

But  $\frac{P_1 - P_2}{\rho g}$  is the difference of pressure heads at section 1 and 2 and it is equal to h

Or  $\frac{P_1 - P_2}{\rho g} = h$  Substituting the value of  $\frac{P_1 - P_2}{\rho g}$  in the above equation, we

Get:

$$h = \frac{V_2^2 - V_1^2}{2g} \quad (1) \quad \checkmark$$

Now applying continuity equation between section 1 and 2

$$a_1 v_1 = a_2 v_2 \quad \text{Or} \quad v_1 = \frac{a_2 v_2}{a_1}$$

Substituting this value of  $v_1$  in equation (1)

$$h = \frac{V_2^2}{2g} - \left( \frac{\frac{a_2 v_2}{a_1}}{2g} \right)^2 = \frac{V_2^2}{2g} \left[ 1 - \frac{a_2^2}{a_1^2} \right] = \frac{V_2^2}{2g} \left[ \frac{a_1^2 - a_2^2}{a_1^2} \right]$$

$$v_2^2 = 2gh \frac{a_1^2}{a_1^2 - a_2^2}$$

$$v_2 = \sqrt{2gh \frac{a_1^2}{a_1^2 - a_2^2}} = \frac{a_1}{\sqrt{a_1^2 - a_2^2}} \sqrt{2gh}$$

Since Discharge is given by  $Q = a_2 v_2$

$$Q = a_2 \frac{a_1}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh} = \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh} \quad (2) \quad \checkmark$$



Equating (2) gives the discharge under ideal conditions and is called, theoretical discharge.

Actual discharge will be less than theoretical discharge.

$$Q_{act} = c_d \times \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \times \sqrt{2gh} \text{--- (3) } \checkmark$$

Where  $c_d$  is co-efficient of Venturimeter and its value is less than 1.

### Value of 'h' given by differential U-tube manometer.

**Case I.** Let the differential manometer contains a liquid which is heavier than the liquid flowing through the pipe.

Let  $s_h$  specific gravity of heavier liquid

$s_o$  specific gravity of liquid flowing through pipe.

X= Difference of the heavier liquid column in U-tube Then  $h = \left[ \frac{s_h}{s_h} \right] \text{--- (4)}$

**Case II.** If the differential manometer contains a liquid which is lighter than the liquid flowing through the pipe, the value of h is given by  $h = \left[ 1 - \frac{s_l}{s_o} \right]$

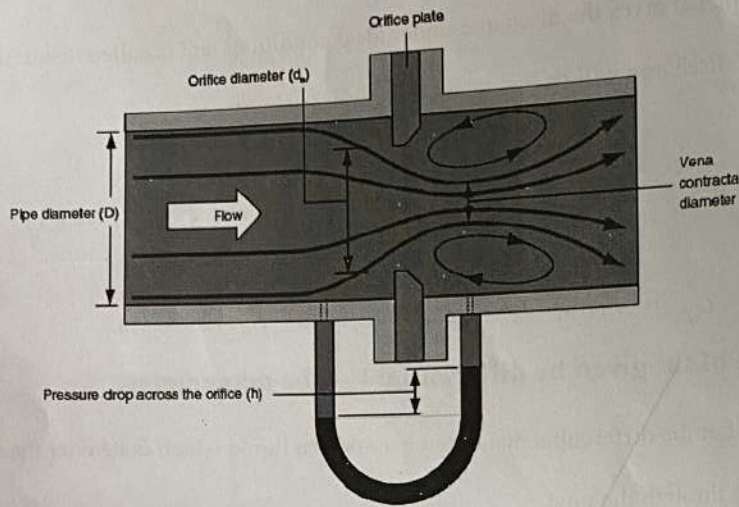
where  $s_l$  specific gravity of lighter liquid in U tube

$s_o$  specific gravity of fluid flowing through the pipe

X= Difference of the heavier liquid columns in U-tube

### Orifice meter or Orifice plate

It is a device used for measuring the rate of flow of a liquid through a pipe. It is a cheaper device as compared to Venturimeter. It also works on the same principal as that of Venturimeter. It consists of a flat circular plate which has a circular sharp edged hole called orifice, which is concentric with the pipe. The orifice diameter is kept generally 0.5 times the diameter of the pipe. Through it may vary from 0.4 to 0.8 times the pipe diameter



A differential manometer is connected at section (1), which is at a distance of about 1.5 to 2.0 times the pipe diameter upstream from the orifice plate, and at section (2), which is at a distance of about half the diameter of the orifice on the downstream side from the orifice plate. Let  $P_1$  = pressure at section (1),  $v_1$  = velocity at section (1),  $a$  = area of pipe at section (1) and  $p_2$ ,  $v_2$ ,  $a_2$ , are corresponding values at section (2). Applying Bernoulli's equation at sections (1) and (2). Applying Bernoulli's Principle we get

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

$$\text{Or } \left( \frac{P_1}{\rho g} + Z_1 \right) - \left( \frac{P_2}{\rho g} + Z_2 \right) = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

$$\text{But Differential head} = h = \left( \frac{P_1}{\rho g} + Z_1 \right) - \left( \frac{P_2}{\rho g} + Z_2 \right)$$

$$h = \frac{V_2^2}{2g} - \frac{V_1^2}{2g} \quad \text{Or } 2gh = V_2^2 - V_1^2$$

Or

$$V_2 = \sqrt{2gh + V_1^2} \quad \text{--- (i)}$$



Now section (2) is at the vena contracta and  $a_2$  represents the area at the vena contracta. If  $a_o$  is the area of orifice then, we have  $C_c = \frac{a_2}{a_o}$  Where  $C_c$  = Co-efficient of contraction  $a_2 = a_o \times C_c$

By continuity equation, we have  $a_1 v_1 = a_2 v_2$  Or  $v_1 = \frac{a_2}{a_1} v_2 = \frac{a_o C_c}{a_1} v_2$  ✓

Substituting the values of  $v_1$  in equation (i), we get

$$v_2 = \sqrt{2gh + \frac{a_o^2 C_c^2 v_2^2}{a_1^2}}$$

Or

$$v_2^2 = 2gh + \left(\frac{a_o}{a_1}\right)^2 C_c^2 v_2^2 \times \left[1 - \left(\frac{a_o}{a_1}\right)^2 C_c^2\right] = 2gh$$

$$v_2 = \frac{\sqrt{2gh}}{\sqrt{1 - \left(\frac{a_o}{a_1}\right)^2 C_c^2}} \quad \checkmark$$

The discharge  $Q = v_2 \times a_2 = v_2 \times a_o C_c a_2 = a_o C_c \frac{a_o C_c \sqrt{2gh}}{\sqrt{1 - \left(\frac{a_o}{a_1}\right)^2 C_c^2}} \dots (iv)$

The above expression is simplified by using  $C_d = C_c \frac{\sqrt{1 - \left(\frac{a_o}{a_1}\right)^2}}{\sqrt{1 - \left(\frac{a_o}{a_1}\right)^2 C_c^2}}$  ✓

Hence we get  $C_c = C_d \frac{\sqrt{1 - \left(\frac{a_o}{a_1}\right)^2 C_c^2}}{\sqrt{1 - \left(\frac{a_o}{a_1}\right)^2}}$

Substituting this value of  $C_c$  in equation (iv). We get

$$Q = a_o \times C_d \frac{\sqrt{1 - \left(\frac{a_o}{a_1}\right)^2 C_c^2}}{\sqrt{1 - \left(\frac{a_o}{a_1}\right)^2}} \times \frac{\sqrt{2gh}}{\sqrt{1 - \left(\frac{a_o}{a_1}\right)^2 C_c^2}}$$

$$\frac{C_d a_o \sqrt{2gh}}{\sqrt{1 - \left(\frac{a_o}{a_1}\right)^2}} = \frac{C_d a_o a_1 \sqrt{2gh}}{\sqrt{a_1^2 - a_o^2}} \quad \checkmark$$



Where  $C_d$  = Co-efficient of discharge for orifice meter.

The co-efficient of discharge for orifice meter is much smaller than that for a Venturimeter.

## NOTCHES

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